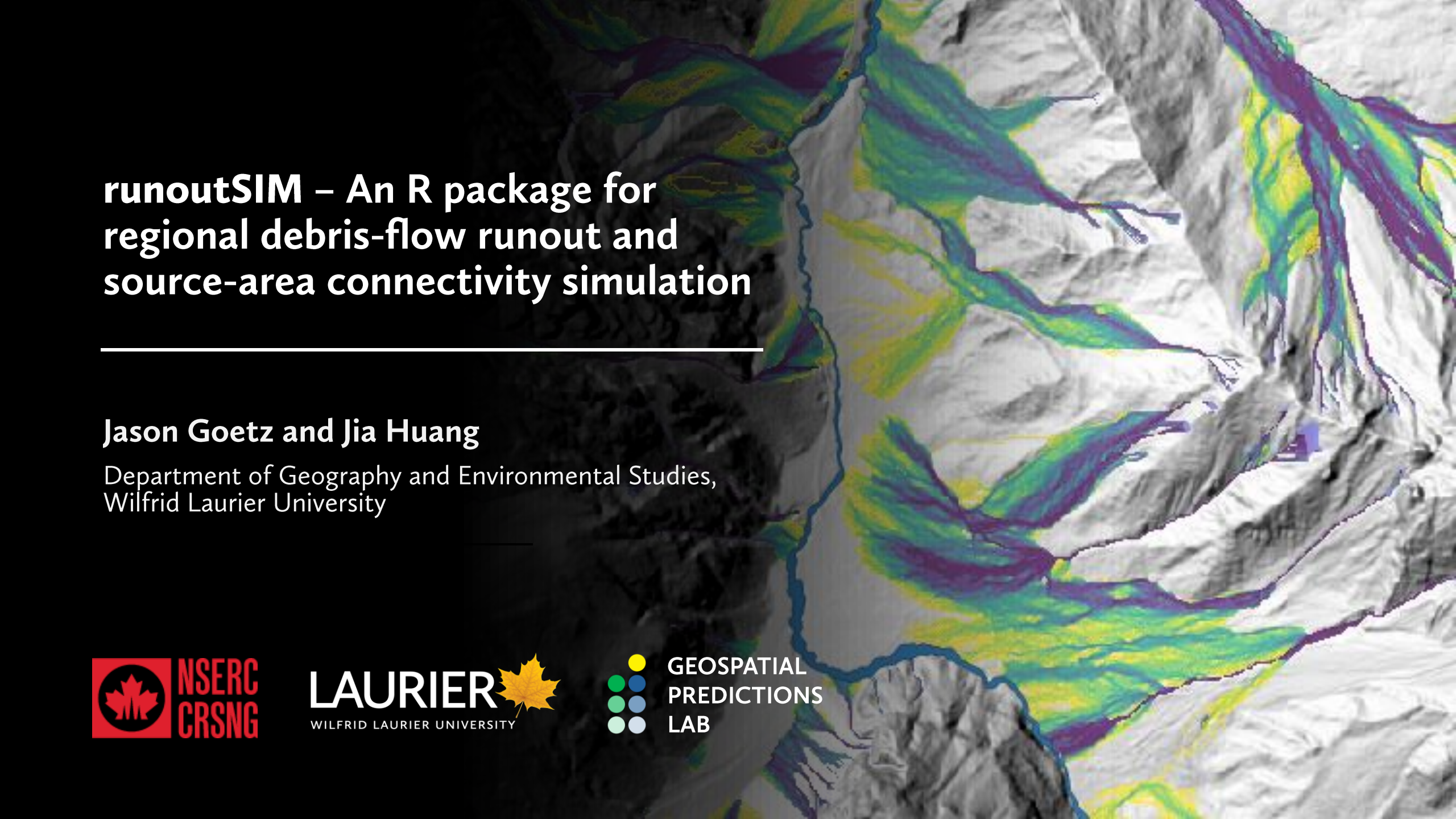
runoutSIM – An R package for regional debris-flow runout and source-area connectivity simulation

Jason Goetz and Jia Huang

Department of Geography and Environmental Studies,
Wilfrid Laurier University



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Why an R package for regional-scale landslide runout simulation?

Enabling advances in modelling

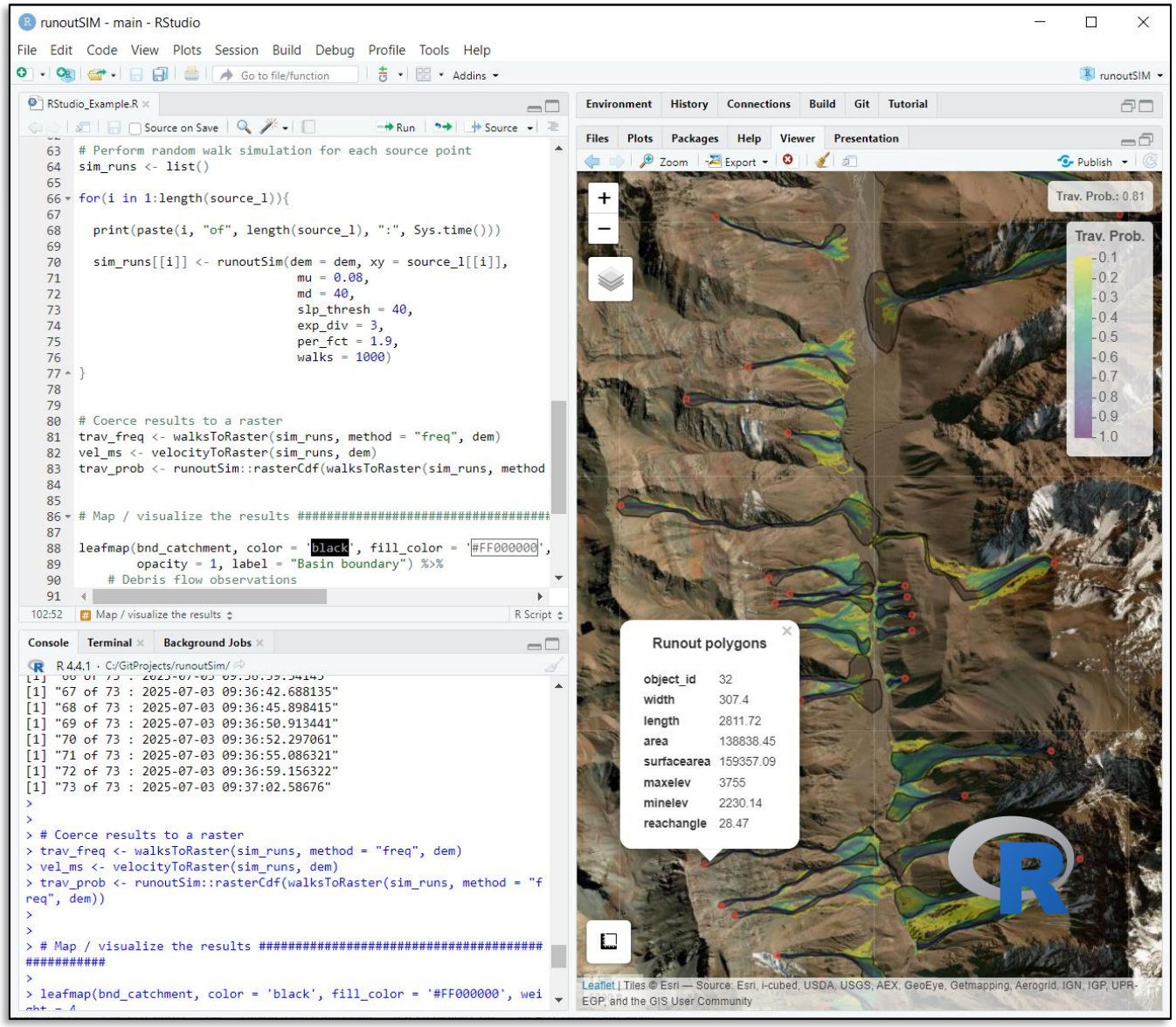
Direct integration with statistical, machine learning, and data visualization tools.

Simplifying workflows

Reducing need for coupling with external software for calibration, performance assessments, and source area modelling.

Regional hazard assessments

Improving the understanding of likely-downstream impacts of debris flows for large-scale extreme precipitation events.



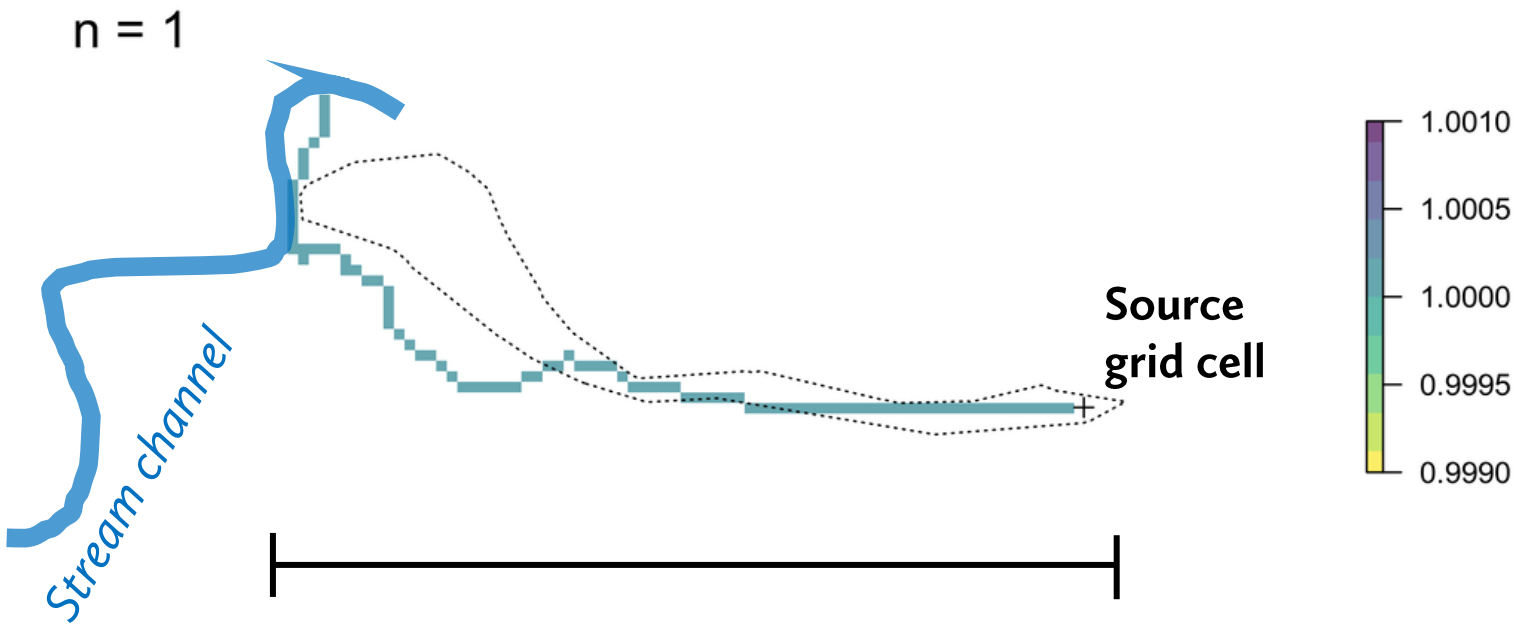
How does runoutSIM work?

An adaptation of the Gravitational Process Path (GPP) model components (Wichman 2017)

+ **source area connectivity probability**

1

Random walk model with 3-parameters to determine runout path / dispersion (Gamma 2000).



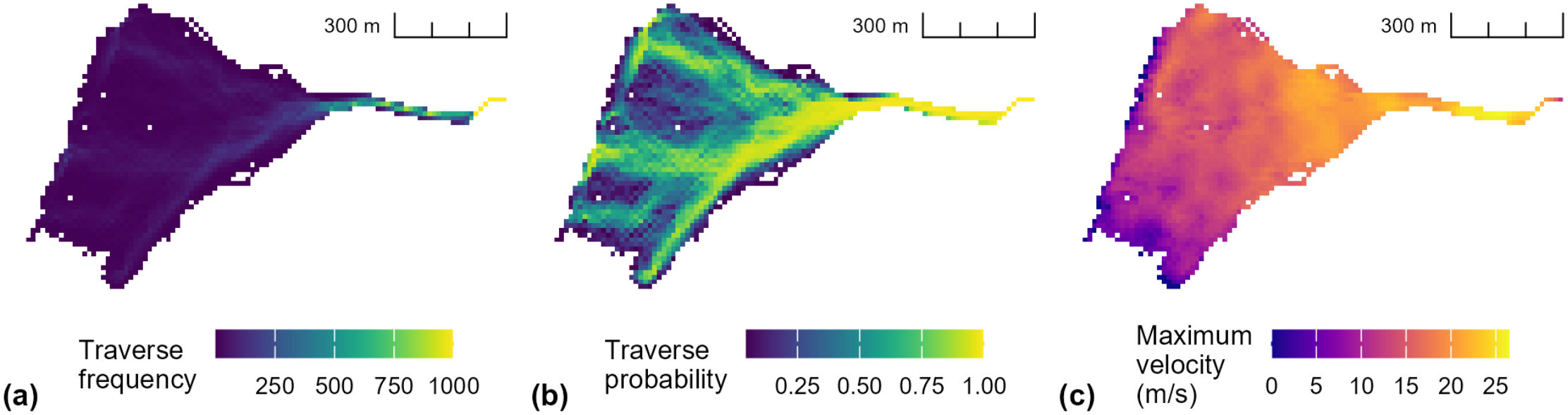
3

Connectivity probability
For each source-area calc. number of intersections per number of walks.

2

Two-parameter friction model controls runout length (Perla et al 1980).

Simulation output example for single source cell



Sum from n=1000 walks (Monte Carlo simulations)

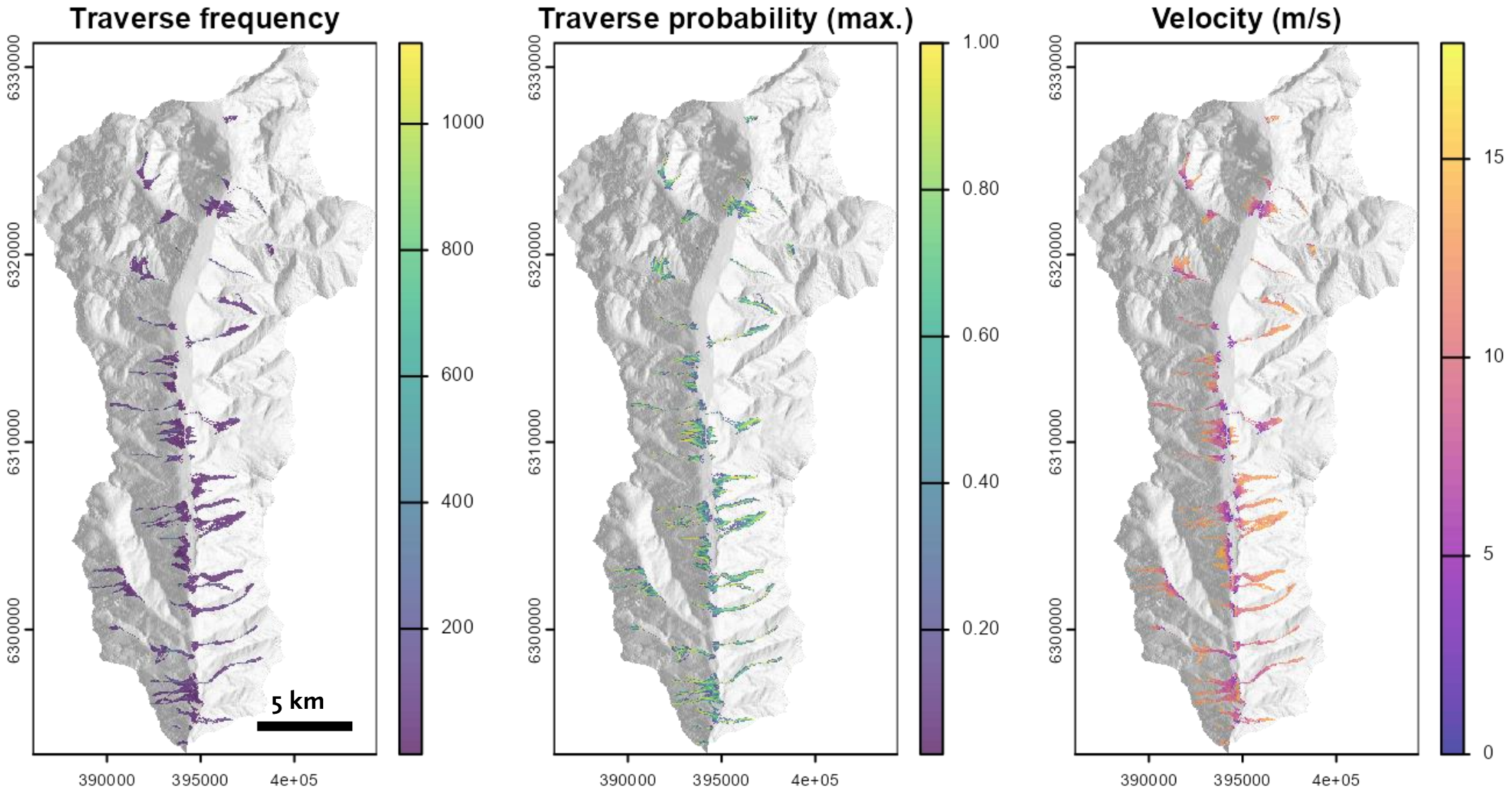
<u>Relative</u> probabilities from an empirical cumulative distribution function

From the two-parameter friction model

Simulation output for multi-source cells

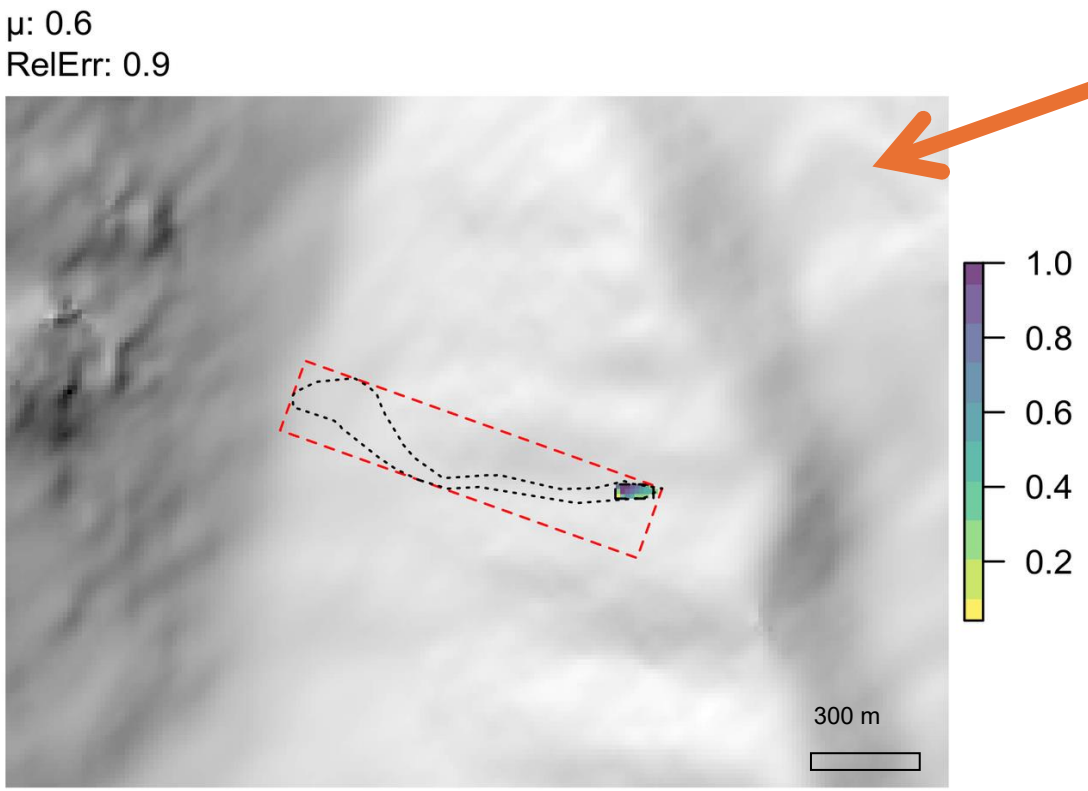
Aggregated by max. relative prob.

Aggregated by max. velocity



Optimizing runoutSIM parameters | customization within R

E.g. Minimize relative error for input runout tracks (Goetz et 2021)



Min. area bounding box

Observed runout length

Estimated runout length

Using existing methods (runoptGPP)

Grid search optimization using min. area bounding boxes

Build your own cost functions

- Distance to impact features
- Center-line-distance
- Connectivity to feature

Pick a suitable optimization method

- Grid search
- Random grid search
- Bayesian optimization

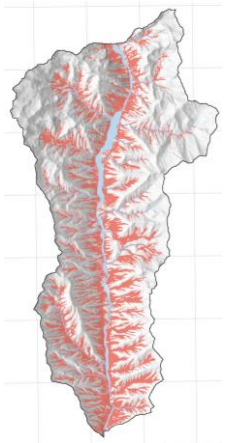
A typical modelling workflow | A case study for Río Olivares basin, Chilean Andes

1A

Source area prediction and classification

Logistic GAM trained on 500 initiation points using slope, curvature, TWI and catchment area.

Classified source area with prob. > 0.5.

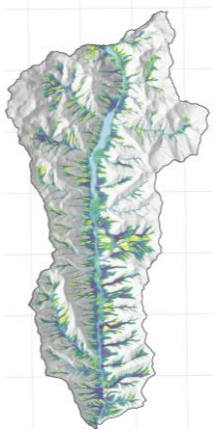


Classified source cells

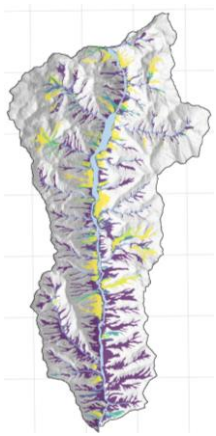
2

Regional simulation

~570 000 source cells (12.5 m resolution) * 1000 walks (~16 hours on 30-thread workstation)



Traverse prob.

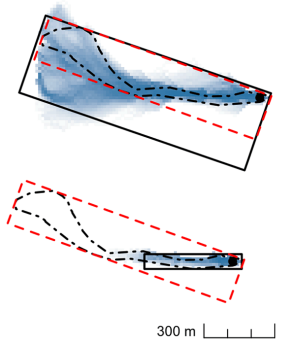


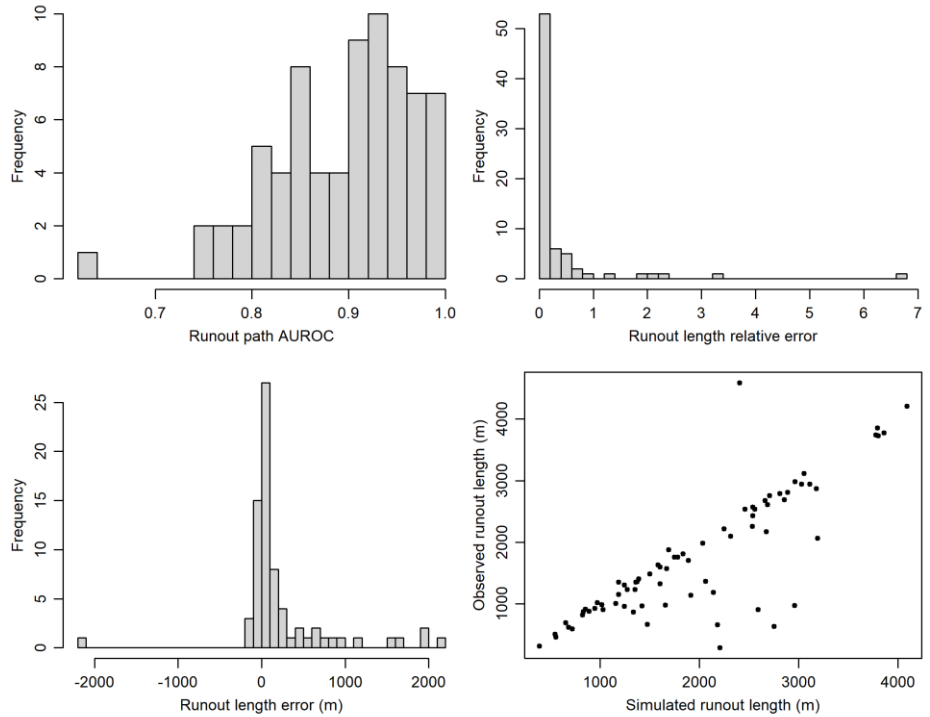
Connectivity prob.

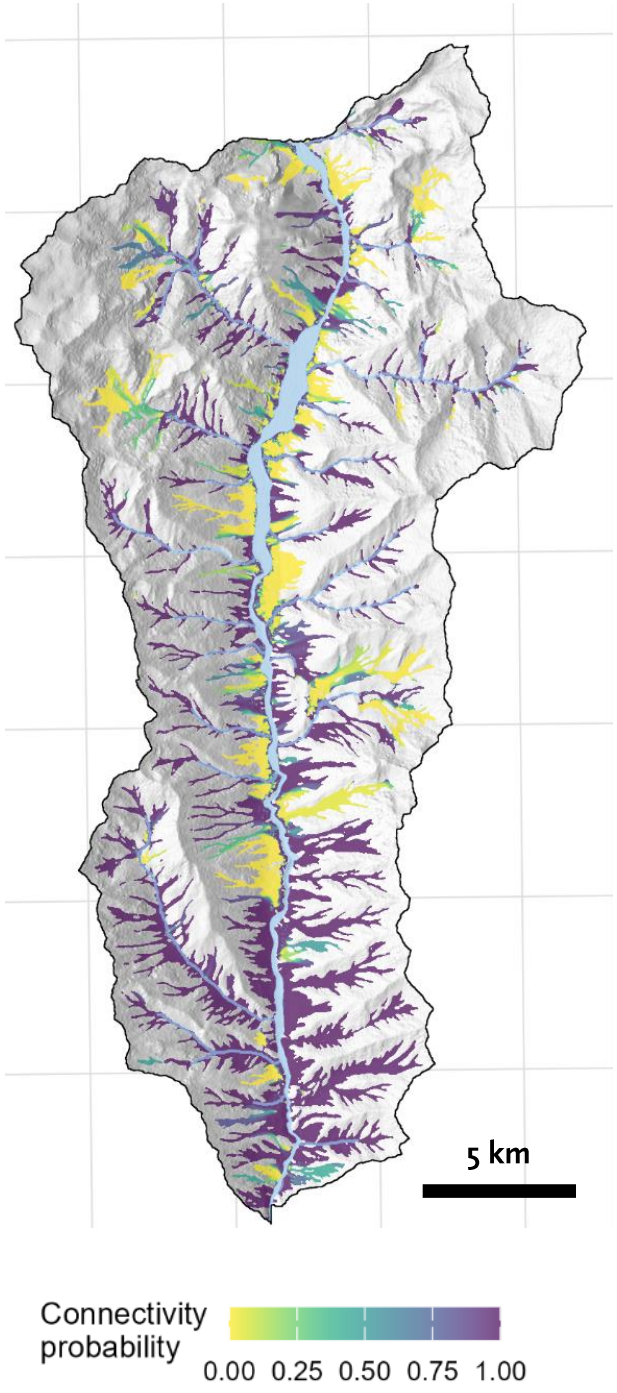
1B

Runout parameter optimization

Random grid search optimizing of 5 parameters (73 observations) using **runoptGPP** package.







A closer look at source area connectivity. Why it matters

Rivers and streams

Identify sediment connectivity.

Infrastructure

Which hillslopes threaten buildings, roads, etc.

Scenario building

Simulate how changed friction (e.g. deforestation) affects connectivity patterns.

Weighted spatial probabilities

Reduce overprediction by accounting for source initiation probability.

runoutSIM | a sandbox environment for regional runout model development

- **Change the code!** an accessible language (R) for model customization (random walk, distance model)
- **Stream-line workflow** for coupling of source area prediction and runout modelling
- **Fit-for-purpose optimization** using existing R functions or build your own
- **Easy-to-start** using tutorials and case study examples
  - A basic introduction
  - Parallel computing
  - Optimizing parameters with runoptGPP

